

No. of Functional Features Included in the Solution

Date	27 June 2026
Team ID	
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

The Smart Lender – Loan Approval Prediction System includes the essential functional features required to perform accurate loan eligibility prediction. Each feature contributes to efficient data processing, machine learning-based prediction, and a smooth user experience through a web-based interface.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input	Accepts applicant details through a web form.	Frontend	Done	High
2	Data Validation	Validates user input before prediction.	Backend	Done	High
3	Loan Prediction	Predicts loan approval using Random Forest.	ML Model	Done	High
4	Result Display	Displays loan approval or rejection instantly.	Frontend	Done	High
5	Model Loading	Loads the trained model for prediction.	Backend	Done	Medium
6	Data Preprocessing	Handles missing values and encodes data.	ML Pipeline	Done	High
7	Model Training	Trains and evaluates ML algorithms.	ML Pipeline	Done	High
8	Cloud Deployment	Deploys the application on Render.	Deployment	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Result Display
2	Backend / Logic	3	Validation, Model Loading, Prediction
3	Database / Storage	2	CSV Dataset, Pickle Model
4	API / Integration	1	Flask Request Handling
5	Security / Authentication	0	Not Implemented (Public App)

Overall Description :-

The Smart Lender – Loan Approval Prediction System successfully implements all the planned functional features required for accurate loan eligibility prediction. The application integrates data preprocessing, machine learning, and a Flask-based web interface to deliver reliable prediction results. All core functionalities have been implemented, tested, and verified, while cloud deployment ensures easy accessibility. The completed solution effectively meets the project objectives and provides a practical, efficient, and user-friendly loan prediction system.